Sports Science Research Update

February 2023

Ability to return to sports after early lateral ligament repair of the ankle in 147 elite athletes. Knee Surgery, Sports Traumatology, Arthroscopy, Ahead of Print: 1-7, 2022.

Acute supplementation with cannabidiol does not attenuate inflammation or improve measures of performance following strenuous exercise. Healthcare, 10(1133): 1-18, 2022.

Air quality is predictive of mistakes in professional baseball and American football. International Journal of Environmental Research and Public Health, 20(542): 1-11, 2023.

Changing the criteria applied to acceleration and deceleration efforts changes the types of player actions detected. Science and Medicine in Football, Ahead of Print: 1-9, 2022.

Effects of cold-water immersion compared with other recovery modalities on athletic performance following acute strenuous exercise in physically active participants: A systematic review, meta-analysis, and meta-regression. Sports Medicine, Ahead of Print: 1-19, 2022.

False Start? An analysis of NFL penalties with and without crowds. Journal of Sports Economics, Ahead of Print: 1-22, 2023.

Hamstring injury rates have increased during recent seasons and now constitute 24% of all injuries in men's professional football: The UEFA Elite Club Injury Study from 2001-02 to 2021-22. British Journal of Sports Medicine, Ahead of Print: 1-7, 2022.

Injury rates remained elevated in the second National Football League season after the onset of the COVID-19 pandemic. Arthroscopy, Sports Medicine, and Rehabilitation, Ahead of Print: 1-13, 2022.

Measuring head acceleration like a CHAMP. Journal of Athletic Training, Ahead of Print: 1-7, 2022.

Muscle architecture, morphology, and mechanical and functional properties of biceps femoris long head in professional soccer players with a prior healed injured hamstring. Journal of Clinical Medicine, 11(7222): 1-10, 2022.

Sports Science Research Update

Planning the microcycle in elite football: To rest or not to rest? International Journal of Sports Physiology and Performance, Ahead of Print: 1-7, 2023.

Post-game recovery of isometric hamstring strength in a high-risk hamstring strain injury group throughout an Australian Football League season. Science and Medicine in Football, Ahead of Print: 1-25, 2022.

Relating American football age of first exposure to patient-reported outcomes and medical diagnoses among former National Football League players: An NFL-LONG study. Sports Medicine, Ahead of Print: 1-12, 2022.

Selecting metrics that matter: Comparing the use of the countermovement jump for performance profiling neuromuscular fatigue monitoring, and injury rehabilitation testing. Strength and Conditioning Journal, Ahead of Print: 1-9, 2023.

Seven points for athletes to consider before using a dietary supplement. AMA Journal of Ethics, 24(5): e443-451, 2022.

Should we lose sleep over sleep disturbances after sports-related concussion? A scoping review of the literature. Journal of Head Trauma Rehabilitation, 37(3): e206-219, 2022.

The current state of concussion knowledge and attitudes in British *American* football. The Physician and Sportsmedicine, Ahead of Print: 1-11, 2022.

The effect of exercise-induced fatigue and heat exposure on soccer-specific decision-making during high-intensity intermittent exercise. PLOS ONE: Dec. 15: 1-24, 2022.

The effects of fixture congestion on injury in professional male soccer: A systematic review. Sports Medicine, Ahead of Print: 1-19, 2022.

The "Worst-Case Scenario": Recovery between repeated high-intensity efforts in Rugby League match-play. Sports Health, Ahead of Print: 1-6, 2022.

Sports Science Research Update

Ability to return to sports after early lateral ligament repair of the ankle in 147 elite athletes. Knee Surgery, Sports Traumatology, Arthroscopy, Ahead of Print: 1-7, 2022.

Purpose: The literature on elite athletes' ability to return to sports (RTS) after early lateral ligament repair of the ankle remains inadequate. The time needed to RTS after modified Broström repair for acute grade III lateral ligament injuries in a large cohort of elite athletes was evaluated while assessing the impact of other associated injuries.

Methods: Prospective data from 147 elite athletes who underwent a primary lateral ligament repair for acute grade III injuries with clinical ankle instability from 2015 to 2019 with a minimum of 2-year follow-up were reviewed. Pertinent details such as characteristics of injury, time taken to RTS, impact of associated injuries on RTS and complications were recorded and analyzed.

Results: The average age was 24.4 years (S.D=4.9) with 89.1% males. Approximately two-thirds of the elite athletes were football players (66.0%) and 21.1% were rugby players. Isolated lateral ligament complex injuries were sustained by 122 (83%) patients, while 25 (17%) had associated injuries (osteochondral lesions of the talus/deltoid injury/syndesmosis injury). All 147 athletes returned to their preinjury level of sports with a median time of 69 days (range 58–132 days). There was no significant difference found in time to RTS between gender, age, and types of sports. However, there was a statistically significant difference noted in the time taken to RTS earlier in patients with isolated lateral ligament injuries when compared to those with associated injuries (68.6 vs. 82.8 days; p=0.004). Multiple conditional linear regression yielded three independent predictors of longer time taken to RTS which were associated deltoid injury (7.5 days longer; 95% CI 2.9–12.3; p=0.002), associated syndesmosis injury (6.7 days longer; 95% CI 0.5–12.8; p=0.034) and associated OLT (30.3 days longer; 95% CI 25.1–35.5; p=0.000).

Conclusion: Early surgical repair for acute grade III lateral ligament injuries in elite athletes yielded excellent rate of RTS at 9–10 weeks and a re-injury rate of 2%. Concomitant injuries will delay RTS after surgical intervention. These findings serve as important guide to managing elite athletes following acute rupture of the lateral ligament complex.

Level of evidence: Level III.

Sports Science Research Update

Acute supplementation with cannabidiol does not attenuate inflammation or improve measures of performance following strenuous exercise. Healthcare, 10(1133): 1-18, 2022.

Abstract - Supplementation with cannabidiol (CBD) may expedite recovery when consumed after exercise. The purpose of this study was to determine if supplementation with CBD reduces inflammation and enhances performance following strenuous eccentric exercise in collegiate athletes. Twenty-four well-trained females (age = 21.2 ± 1.8 years, height = 166.4 ± 8 cm, weight = 64.9 ± 9.1 kg) completed 100 repetitions of unilateral eccentric leg extension to induce muscle damage. In this crossover design, participants were randomized to receive 5 mg/kg of CBD in pill form or a placebo 2 h prior to, immediately following, and 10 h following muscle damage. Blood was collected, and performance and fatigue were measured prior to, and 4 h, 24 h, and 48 h following the muscle damage. Approximately 28 days separated treatment administration to control for the menstrual cycle. No significant differences were observed between the treatments for inflammation, muscle damage, or subjective fatigue. Peak torque at 60◦/s ($p = 0.001$) and peak isometric torque ($p = 0.02$) were significantly lower 24 h following muscle damage, but no difference in performance was observed between treatments at any timepoint. Cannabidiol supplementation was unable to reduce fatigue, limit inflammation, or restore performance in well-trained female athletes.

Sports Science Research Update

Air quality is predictive of mistakes in professional baseball and American football. International Journal of Environmental Research and Public Health, 20(542): 1-11, 2023.

Abstract - Air quality is a growing environmental concern that has implications for human physical and mental health. While air pollution has been linked to cognitive disease progression and declines in overall health, the impacts of air quality on athletic performance have not been extensively investigated. Much of the previous research focused on endurance sports indicates that air quality negatively impacts athletic performance; however, the effects of air quality on non-endurance elite team performance remains largely unknown. The purpose of this study was to examine the impact of air quality on errors committed by Major League Baseball (MLB) teams, interceptions thrown by quarterbacks in the National Football League (NFL), and overall quarterback performance in the NFL. Linear regression analysis was used to determine the impact of the median air quality index (AQI) of counties with MLB and NFL teams on errors, interceptions, and overall quarterback performance of players on those MLB and NFL teams. AQI was a significant positive predictor of errors and interceptions, indicating increased errors and interceptions with decreased air quality. Similarly, quarterback performance was significantly reduced for quarterbacks from teams in counties with worse air quality. These findings suggest that air quality has a significant impact on performance in the MLB and NFL, indicating impairments in physical and cognitive performance in professional athletes when competing in areas with poorer air quality. Hence, it is likely that air quality impacts athletic performance in numerous sports that have not yet been investigated.

Sports Science Research Update

Changing the criteria applied to acceleration and deceleration efforts changes the types of player actions detected. Science and Medicine in Football, Ahead of Print: 1-9, 2022.

Abstract - This study identified and profiled the types of player actions of Australian football players executed during high acceleration and/or deceleration efforts and identified how the duration exceeding the acceleration or deceleration threshold influenced the identified executed player actions. The commencing speed of the acceleration and deceleration efforts were also assessed. Acceleration and deceleration efforts of 22 Australian football players monitored with a 10-Hz GNSS device across two matches were analyzed. Synchronized video footage of the matches was used to identify the player actions executed during acceleration and deceleration efforts. Linear mixed models were conducted to examine the relationship between commencing speed, player action, and duration exceeding the effort threshold. 'Linear movement' and 'jump' were the most frequently executed player actions for high accelerations and 'stop' and 'being tackled' for high decelerations. Significant differences were observed in commencing speed between player actions for the high deceleration efforts. Furthermore, it was found that a higher commencing speed was associated with an increased duration exceeding the deceleration threshold. These findings will help practitioners and researchers with selecting minimum-duration criteria for acceleration and deceleration efforts and provide an understanding of the types of player actions executed during these efforts.

Practical application

● Practitioners and researchers should be aware that selecting a minimum-duration above the threshold for detecting high acceleration and deceleration efforts will influence the corresponding type of player actions detected during these efforts.

● Practitioners and researchers can use the findings of this research to help with selecting minimum-duration criteria that best resemble the player's actions of interest and exclude movements they are not interested in.

● It is recommended that training practices for high acceleration and deceleration efforts replicate the player actions associated with these efforts evident in this research, over various commencing speeds.

Sports Science Research Update

Effects of cold-water immersion compared with other recovery modalities on athletic performance following acute strenuous exercise in physically active participants: A systematic review, meta-analysis, and meta-regression. Sports Medicine, Ahead of Print: 1-19, 2022.

Background: Studies investigating the effects of common recovery modalities following acute strenuous exercise have reported mixed results.

Objectives: This systematic review with meta-analysis and meta-regression compared the effects of cold-water immersion (CWI) against other common recovery modalities on recovery of athletic performance, perceptual outcomes, and creatine kinase (CK) following acute strenuous exercise in physically active populations.

Study Design: Systematic review, meta-analysis, and meta-regression.

Methods: The MEDLINE, SPORTDiscus, Scopus, Web of Science, Cochrane Library, EmCare, and Embase databases were searched up until September 2022. Studies were included if they were peer reviewed, published in English, included participants who were involved in sport or deemed physically active, compared CWI with other recovery modalities following an acute bout of strenuous exercise, and included measures of performance, perceptual measures of recovery, or CK.

Results: Twenty-eight studies were meta-analyzed. CWI was superior to other recovery methods for recovering from muscle soreness, and similar to other methods for recovery of muscular power and flexibility. CWI was more effective than active recovery, contrast water therapy and warm-water immersion for most recovery outcomes. Air cryotherapy was significantly more effective than CWI for the promotion of recovery of muscular strength and the immediate recovery of muscular power (1-h post-exercise). Meta-regression revealed that water temperature and exposure duration were rarely exposure moderators.

Conclusion: CWI is effective for promoting recovery from acute strenuous exercise in physically active populations compared with other common recovery methods.

Sports Science Research Update

Key Points

- Cold-water immersion (CWI) was more effective than active recovery, contrast water therapy and warm-water immersion for most outcomes, including reducing muscle soreness and improving muscular power.
- Water temperature and exposure duration were rarely impactful effect moderators; however, there was a dose–response effect of a lower temperature and shorter duration positively influencing the recovery of muscular power after CWI 24-h post-exercise when compared with active recovery.
- Air cryotherapy was more effective than CWI for immediately recovering muscular power (1-h post-exercise) and for recovering muscular strength.

Sports Science Research Update

False Start? An analysis of NFL penalties with and without crowds. Journal of Sports Economics, Ahead of Print: 1-22, 2023.

Abstract - The lack of crowds at sports fixtures as a result of COVID-19 restrictions has allowed researchers a unique opportunity to examine the widely accepted convention of home advantage. This work takes a slightly different approach, by examining within game outcomes. Specifically using play-by-play data from the NFL, this paper asks whether the lack of crowds aided a Quarterback's ability to manipulate opposition defenses. Results suggest this was the case, though effects are not uniform by home and away team.

Sports Science Research Update

Hamstring injury rates have increased during recent seasons and now constitute 24% of all injuries in men's professional football: The UEFA Elite Club Injury Study from 2001-02 to 2021-22. British Journal of Sports Medicine, Ahead of Print: 1-7, 2022.

Objectives: To: (1) describe hamstring injury incidence and burden in male professional football players over 21 seasons (2001/02 to 2021/22); (2) analyze the time-trends of hamstring muscle injuries over the most recent eight seasons (2014/15 to 2021/22); and (3) describe hamstring injury location, mechanism and recurrence rate.

Methods: 3909 players from 54 teams (in 20 European countries) from 2001/02 to 2021/22 (21 consecutive seasons) were included. Team medical staff recorded individual player exposure and time-loss injuries. Time-trend analyses were performed with Poisson regression using generalized linear models.

Results: 2636 hamstring injuries represented 19% of all reported injuries, with the proportion of all injuries increasing from 12% during the first season to 24% in the most recent season. During that same period, the percentage of all injury absence days caused by hamstring injuries increased from 10% to 20%. Between 2014/15 and 2021/22, training hamstring injury incidence increased (6.7% annually, 95% CI 1.7% to 12.5%) as did burden (9.0% annually, 95% CI 1.2% to 18.3%). During those years, the match hamstring injury incidence also increased (3.9% annually, 95% CI 0.1% to 7.9%) and with the same trend (not statistically significant) for match hamstring injury burden (6.2% annually, 95% CI −0.5% to 15.0%).

Conclusions: Hamstring injury proportions—in number of injuries and total absence days—doubled during the 21-year period of study. During the last eight seasons, hamstring injury rates have increased both in training and match play.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Hamstring muscle injuries increased in incidence in men's professional football from 2001 to 2014.

⇒ Since then, players train more intensely and their match calendar is more crowded. Many professional teams aim to prevent muscle injury as part of their strength and conditioning programs.

WHAT THIS STUDY ADDS

⇒ During the recent eight seasons (2014/15 to 2021/22), the incidence and burden of hamstring injuries during training and match play have increased significantly.

Sports Science Research Update

⇒ The proportion of injuries diagnosed as hamstring injuries increased from 12% in 2001/02 to 24% in 2021/22. ⇒ The proportion of all injury absence days caused by hamstring injuries has increased from 10% in 2001/02 to 20% in 2021/22.

⇒ Around 18% of all reported hamstring injuries were recurrences with over two-thirds occurring within 2months of the footballer's return to play.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ These data on incidence and burden of hamstring injuries provide a strong rationale for teams to keep focusing on preventing first and recurrent hamstring injuries.

⇒ The high rate of recurrent hamstring injury within 2months of return to play suggests this period is a particularly important time for team clinicians to: (1) carefully monitor players completing evidence-based rehabilitation; (2) manage training and match loads; and (3) maintain preventive programs.

Sports Science Research Update

Injury rates remained elevated in the second National Football League season after the onset of the COVID-19 pandemic. Arthroscopy, Sports Medicine, and Rehabilitation, Ahead of Print: 1-13, 2022.

Purpose: The purpose of this study was to compare the injury incidence of the 2018-2019 and 2020 National Football League (NFL) seasons with the 2021 season.

Methods: Publicly released NFL weekly injury reports were queried to identify players listed as "out" or placed on injured reserve (IR) for at least 1 game in the 2018-2021 seasons. Injuries were then categorized into upper extremity, lower extremity, spine/core, and head. Incidence per 1,000 athlete exposures were calculated for each season, and proportions of injuries by position were calculated separately for the 2018-2019, 2020, and 2021 cohorts. Incidence rate ratios (IRR) were used to compare injury rates.

Results: Overall injury incidence in the 2021 NFL season increased compared to the pre-COVID-19 seasons (2018-2019) in all anatomical zones except for the upper extremity (28.70 vs 23.09 per 1,000 exposures, IRR 1.24 [95% CI: 1.14-1.36]; $P < .001$). The injury rate remained elevated and further increased in 2021 compared to the 2020 season for all anatomical zones other than the spine/core [28.70 vs 21.64 per 1,000 exposures, IRR 1.33 (1.19-1.47); $P < .001$]. No significant difference existed during the early season (weeks 1-4); however, injury rates after week 4 increased in 2021 compared to both the 2018-2019 and 2020 seasons.

Conclusion: The injury incidence in the 2021 season remained elevated and increased further compared to both the 2018-2019 and 2020 seasons. Traumatic injuries resulting in missed games increased despite return to a more traditional season since the beginning of the COVID-19 pandemic. The injury rates significantly increased in mid-season to late season.

Sports Science Research Update

Measuring head acceleration like a CHAMP. Journal of Athletic Training, Ahead of Print: 1-7, 2022.

Overview - To this end, the Consensus Head Acceleration Measurement Practices (CHAMP) group was founded to develop and recommend best practices for the collection, analysis, and reporting of head acceleration measurement data in sport. Comprised of scientists and clinicians from a variety of backgrounds (including industry representation), the group undertook a comprehensive effort to define current best practices in head kinematic measurement, culminating in a series of manuscripts outlining consensus methodologies. CHAMP prioritized 6 areas of focus:

• Study design and statistical analysis in studies of head acceleration measurement • Laboratory validation of wearable head kinematic devices

• On-field validation and use of wearable head kinematic devices

• Video analysis of head acceleration events

• Physical reconstruction of head acceleration events

• Computational modeling of head acceleration events.

Work groups for each focus area, led by experts in the field, drafted consensus statements that outline currently recommended best practices for many aspects of head acceleration measurement. CHAMP workgroups and key stakeholders convened at a consensus conference held in Philadelphia in March 2022. All attendees at the conference participated in an open scientific discussion of the key concepts and then formally voted on each consensus statement. Details of this process and all participants are summarized here.

Sports Science Research Update

Muscle architecture, morphology, and mechanical and functional properties of biceps femoris long head in professional soccer players with a prior healed injured hamstring. Journal of Clinical Medicine, 11(7222): 1-10, 2022.

Objective: The aim of the present study was to compare the fascicle length, pennation angle, muscle thickness and stiffness of the biceps femoris long head, and eccentric hamstring strength between injured dominant limbs, injured non-dominant limbs, uninjured dominant limbs and uninjured non-dominant legs in previously injured players, and between dominant and non-dominant legs in uninjured elite soccer players.

Materials and Methods: Twenty elite soccer players participated in this study. Ultrasound imaging and MyotonPRO were used to determine the morphological and mechanical properties of the biceps femoris long head. Isokinetic and Nordic hamstring exercises were used to assess eccentric hamstring strength.

Results: Previously injured players showed substantially lower fascicle length and muscle thickness, and significantly higher biceps femoris long head stiffness than uninjured players, without differences between limbs.

Conclusion: The morphological and mechanical properties of elite soccer players with hamstring injury history were different from those in uninjured players. The lack of differences between limbs showed that these values are characteristics of individual players that must be considered in the design of programs to prevent BFlh injury.

Sports Science Research Update

Planning the microcycle in elite football: To rest or not to rest? International Journal of Sports Physiology and Performance, Ahead of Print: 1-7, 2023.

Purpose: To examine the association between the programming of days off (ie, no pitch training, days off feet) within turnarounds of varying length and injury rate in elite soccer.

Methods: Retrospective data from 56 team seasons of players belonging to 18 elite teams performing in top leagues including the English Premier League, Italian Serie A, Bundesliga, Scottish Premiership, Major League Soccer, and the Dutch Eredivisie from January 2018 to December 2021 were analyzed (total of 1578 players, 2865 injuries, 2859 noninternational matches, and 12,939 training-session days). The turnarounds examined lasted from 3 to 8 days. Only injuries with ≥3-day time loss were retained for analysis. The authors then looked at the injury rate within each microcycle in relation to the presence of a day off or not and its programming sequences in relation to the previous match (ie, day off at D + 1 vs D + 2 for the day after the match or the following, respectively).

Results: During 3- and 7-day turnarounds, the sequences including the day off feet at D + 2 were associated with 2 to 3 times lower overall noncontact injury rates than the other programming sequences (Cohen d: 0.9–2.7). For the other turnarounds, the differences between the sequences were unclear.

Conclusion: The programming of a day off (or at least "off feet") at D + 2 may be associated with moderately to largely lower incidence of noncontact injuries, especially during 3- and 7-day turnarounds.

Practical Applications: The present study showed for the first time, using a large pool of data from elite football, that while planning a day off (at least off-feet) per se may not share clear associations with injury rate, its programming (ie, when) within the microcycle. In practice, at least for the 3- and 7-day turnarounds examined, programming the (only) day off-feet of the microcycle at D + 2 was associated with 2 to 3 times less overall injury rates than either not having a day off-feet, or programming the latter at D + 1. Future studies should also examine, within each turnaround length, the actual load of each training day in relation to the different day off programming strategies.

Sports Science Research Update

Post-game recovery of isometric hamstring strength in a high-risk hamstring strain injury group throughout an Australian Football League season. Science and Medicine in Football, Ahead of Print: 1-25, 2022.

Abstract - This study sought to investigate post-game hamstring strength recovery of 26 Australian Football League (AFL) players with a previous hamstring strain injury (HSI) across an AFL season. Maximal unilateral isometric knee flexion strength was assessed using an externally fixed dynamometer, and inter-session reliability was measured during the pre-season period. Linear mixed effects models investigated the influence of numerous variables on post-game hamstring strength decrement (relative change between initial weekly test and individual baseline) and individual within-week strength change following gameplay. The test demonstrated good inter-tester reliability (ICC = 0.81–0.88; CV = 6.73–7.33), and an acceptable level of error (MAE = 5.77–7.14%). Player as a random effect strongly influenced post-game strength decrement and within-week strength change (marginal R^2 = 0.185–0.407; conditional R^2 = 0.455–0.654). Within-week hamstring strength change was strongly determined by post-game strength decrement alone (estimate = 0.51, 95% CI = -0.66- -0.36 ; η^2 = 0.32; *P*=<0.001) and in interaction with number of days post-game (estimate = 0.43, 95% CI = 0.20–0.66; η^2 = 0.096; *P*=<0.001). This study shows the importance of early individual assessment of post-game hamstring strength in players with prior hamstring injury and could be valuable to inform post-game hamstring recovery in future applications.

Sports Science Research Update

Relating American football age of first exposure to patient-reported outcomes and medical diagnoses among former National Football League players: An NFL-LONG study. Sports Medicine, Ahead of Print: 1-12, 2022.

Background: The age of first exposure (AFE) to American football participation is a growing concern for late-life function. Mixed evidence exists surrounding AFE and may be attributed to varied methods employed across studies.

Objective: To examine the associations between AFE to American football participation with measures of cognitive, behavioral, and physical function and brain-related medical diagnoses across age categories among former National Football League players.

Methods: We conducted a cross-sectional survey of 1784 former players (age: 52.3±16.3 years, AFE: 11.3±2.9 years, years of football: 17.5±4.5 years, 86.9%≥one lifetime concussion). Players completed a general health questionnaire recording demographics, football playing history (including AFE), and diagnoses (anxiety, depression, any form of dementia, mild cognitive impairment). Players completed Patient-Reported Outcomes Measurement Information System (PROMIS) measures assessing domains of cognitive and physical function, anxiety, depression, sleep disturbance, pain interference, and emotional-behavioral dyscontrol. Multivariable linear and binomial regression models were used to examine the associations of AFE and age with PROMIS outcomes and diagnoses, respectively.

Results: No significant AFE by age interactions were detected for PROMIS outcomes ($p \geq 0.066$) or diagnoses ($p \geq 0.147$). Younger AFE associated with higher PROMIS scores of anxiety ($B = -0.22$, $p = 0.016$), depression ($B = -0.22$, $p = 0.010$), sleep disturbances ($B = -0.16$, $p = 0.007$), pain interference ($B = -0.19$, $p = 0.014$), and emotional-behavioral dyscontrol ($B = -0.22$, $p = 0.019$). Age was associated with all PROMIS outcomes ($p \leq 0.042$). AFE was not associated with the prevalence of anxiety, depression, dementia, or mild cognitive impairment ($p \geq 0.449$), while age was ($p \leq 0.013$).

Conclusions: AFE was significantly associated with PROMIS outcomes, albeit low-strength associations (i.e., effect sizes), but not with diagnoses. Our findings indicate AFE is a significant but minor contributing factor for health-related quality of life in this cohort. Future work should incorporate additional characterizations of cumulative head impacts and related factors when examining long-term outcomes associated with football participation.

Sports Science Research Update

Key Points

- Age of first exposure (AFE) did not demonstrate any significant interaction effects with age when examining late-life function and medical diagnosis prevalence, which suggests any AFE effect does not appear have an aging-dose response.
- AFE was weakly associated with self-reported function and was not associated with medical diagnoses among former NFL players, and age had greater independent associations with self-reported function and diagnoses than AFE.
- Stakeholders (e.g., clinicians, individuals who have played football, family members) may not need to be overly concerned with the age they started playing football and their viewed cognitive and physical function.

Sports Science Research Update

Selecting metrics that matter: Comparing the use of the countermovement jump for performance profiling neuromuscular fatigue monitoring, and injury rehabilitation testing. Strength and Conditioning Journal, Ahead of Print: 1-9, 2023.

Abstract - The countermovement jump (CMJ) is one of the most used performance assessments in strength and conditioning. Although numerous studies discuss the usability of different metrics in this test, this is often performed within the context of a specific aim. However, to our knowledge, no information currently exists providing practitioners with some over-arching recommendations on which metrics to choose when the purpose of using the test differs. This article discusses how the metrics selected to monitor during CMJ testing may differ when aiming to use it as a proxy for athletic performance, as part of neuromuscular fatigue monitoring, or as part of a test battery for return to performance in injured athletes.

PERFORMANCE PROFILING

Purpose: to assess associations with other measures of performance

Metric	What it Assesses
Jump height [cm]	Vertical displacement of the athlete's center of mass
Peak power [W]	Highest rate of doing work during the jump
Mean propulsive force [N]	Mean force produced during ascent
Propulsive impulse [Ns]*	Product of force and time during ascent

Output metrics, including the metrics above, correspond with physical capacities that benefit sport performance, including strength, linear speed, and change-of-direction ability.

NEUROMUSCULAR FATIGUE

Purpose: to detect when neuromuscular fatigue is present

Metric	What it Assesses
Reactive Strength Index (RSI) Modified*	Ratio of output (jump height) to time spent to produce the output (contraction time)
Time to take-off [s]	Total duration from initiation of movement to take-off
Propulsive phase duration [s]	Time spent during ascent (prior to take-off)
Time to peak power [s]*	Amount of time it takes before the greatest amount of power is produced

Time-based metrics, including the metrics above, correspond with acute neuromuscular fatigue (induced by intense physical activity).

RETURN FROM INJURY

Purpose: to assess post-injury progress and preparedness for return to sport

Metric	What it Assesses
Peak propulsive force [N]	Greatest amount of force produced during ascent
Peak landing force [N]	Greatest amount of force produced upon landing from the jump
Landing impulse [Ns]*	Product of force and time during landing from the jump
Asymmetry [%]	Differences in metric outcomes between limbs

Deceleration, landing, and inter-limb asymmetry metrics can differentiate between athletes with and without injuries.

*Ensure all component parts are concurrently monitored to understand what is driving any changes in data over time

Note: It is oftentimes beneficial to evaluate countermovement jump metrics relative to body mass. Above all, the practitioner should be consistent in how they evaluate their data (absolute vs. relative) to appropriately appraise differences between athlete cohorts and changes over time.

Sports Science Research Update

Seven points for athletes to consider before using a dietary supplement. AMA Journal of Ethics, 24(5): e443-451, 2022.

Abstract - Performance-enhancing drugs (PEDs) have been used by athletes for as long as sporting competitions have existed. To protect the health and safety of athletes and promote fair play, banned substance lists were developed that include several classes of PEDs. Evidence shows that a majority of athletes use dietary supplement products to aid their training and support their health. Evidence also indicates that use of some dietary supplements carries a risk because the products may contain banned PEDs. Consumers and athletes should weigh a number of considerations before purchasing and consuming dietary supplements to protect their health, reputation, and the spirit of fair competition.

Sports Science Research Update

Should we lose sleep over sleep disturbances after sports-related concussion? A scoping review of the literature. Journal of Head Trauma Rehabilitation, 37(3): e206-219, 2022.

Objective: A single, severe traumatic brain injury can result in chronic sleep disturbances that can persist several years after the incident. In contrast, it is unclear whether there are sleep disturbances after a sports-related concussion (SRC). Considering growing evidence of links between sleep disturbance and neurodegeneration, this review examined the potential links between diagnosed SRCs and sleep disturbances to provide guidance for future studies.

Methods: The scoping review undertook a systematic search of key online databases (Scopus, MEDLINE, SportDiscus, and Web of Science) using predetermined search terms for any articles that examined sleep after concussion. A screening criterion using agreed inclusion and exclusion criteria was utilized to ensure inclusion of relevant articles.

Design: This scoping review is guided by the PRSIMA Scoping Review report.

Results: Ten studies met the inclusion criteria, reporting on 896 adults who had experienced an SRC. Comparison with 1327 non-SRC adults occurred in 8 studies. Nine studies subjectively examined sleep, of which all but one study reported sleep disturbances after an SRC. Three studies objectively measured sleep, with 2 studies indicating large coefficients of variation of sleep duration, suggesting a range of sleep responses after an SRC. The only study to examine overnight polysomnography showed no differences in sleep metrics between those with and without an SRC. No studies examined interventions to improve sleep outcomes in people with concussion.

Conclusions: This scoping review indicates preliminary evidence of sleep disturbances following an SRC. The heterogeneity of methodology used in the included studies makes consensus on the results difficult. Given the mediating role of sleep in neurodegenerative disorders, further research is needed to identify physiological correlates and pathological mechanisms of sleep disturbances in SRC-related neurodegeneration and whether interventions for sleep problems improve recovery from concussion and reduce the risk of SRC-related neurodegeneration.

Sports Science Research Update

The current state of concussion knowledge and attitudes in British American football. The Physician and Sportsmedicine, Ahead of Print: 1-11, 2022.

Objectives: To examine concussion knowledge and concussion attitudes of players, coaches, and support staff in British American Football (BAF).

Methods: Data from players, coaches and support staff (n = 236) were collected from across all leagues in BAF. An online survey tool was used which included the Rosenbaum Concussion Knowledge and Attitudes Survey (RoCKAS), and questions examining concussion education and perceived risk of participating in football.

Results: The mean score on the RoCKAS concussion knowledge was 21.0 ± 2.1 of a possible score of 25 reflecting good knowledge. Of a possible score of 65, the mean concussion attitude score was 55.6 ± 6.1 showing safe attitude. Whilst an overall safe attitude was seen, almost half of participants (45.3%) noted they would continue to play with a concussion. No relationship was found between CAI and prior concussion history. Fifty seven percent of participants agreed the benefits of playing football out-weighed the risks. Forty eight percent reported that they had received no concussion-related education in the past 12 months.

Conclusion: BAF participants have good concussion knowledge and safe attitudes. However, risky behavior is demonstrated through unsafe likelihood to report and attitude to long-term health risks. Access to the British American Football Association (BAFA) concussion policy and education was poor raising questions over what sources of information stakeholders are drawing their knowledge from. These findings can help form the foundation of educational interventions (e.g. coaching workshops) to challenge current misconceptions and improve likelihood to report concussion in BAF.

Sports Science Research Update

The effect of exercise-induced fatigue and heat exposure on soccer-specific decision-making during high-intensity intermittent exercise. PLOS ONE: Dec. 15: 1-24, 2022.

Abstract - Global warming and the globalization of sport has increased the prevalence of sports competitions being held in hot environments. However, there is currently limited research investigating the impact of the heat on soccer-specific decision-making skills during exercise reflective of the physical demands of match-play. Therefore, the effects of heat exposure on physical and soccer-specific decision-making performance, biological markers (i.e., metanephrines), appraisal (i.e., challenge vs. threat) and affective states, during prolonged high-intensity intermittent exercise were investigated. Nine well-trained male soccer players completed a 92-min cycling intermittent sprint protocol (CISP), whilst simultaneously responding to a series of soccer-specific decision-making trials at various time points, in two temperature conditions: hot (32˚C, 50%rh) and temperate (18˚C, 50%rh). Results showed that decision-making score ($p = .030$) was impaired in the hot compared to the temperate condition. There was a reduced workload in the second half during the hot condition ($p = .016$), which coincided with a heightened threat state ($p = .007$) and more unpleasant feelings ($p = .008$) experienced in the hot, compared to temperate, condition. Furthermore, plasma normetanephrine (NMET) was higher at half-time ($p = .012$) and post-CISP ($p \leq .001$). Also, plasma metanephrine (MET) was higher post-CISP ($p = .009$) in the hot compared to temperate condition, reflecting a heightened stress response. Our findings highlight the need for practitioners to consider the detrimental effects heat exposure can have on both physical and decision-making performance when looking to facilitate performance in hot conditions.

Sports Science Research Update

The effects of fixture congestion on injury in professional male soccer: A systematic review. Sports Medicine, Ahead of Print: 1-19, 2022.

Background: Professional soccer teams are often required to compete with≤4 days recovery between matches. Since congested schedules reduce recovery time between matches, players are possibly at an increased injury risk. To date, there are no published systematic reviews on the impact of match congestion on injuries during professional male soccer.

Objective: The aim of this systematic review was to assess the effects of fixture congestion on injuries during professional soccer. Methods Following pre-registration on the Open Science Framework (https://osf.io/86m25/) and conforming with Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, systematic searches of four electronic databases (PubMed, Scopus, SPORTDiscus, and Web of Science) were conducted by independent researchers from inception until February 2022. Articles were included if they were original articles written in English and contained relevant time-loss injury data (injury that results in unavailability for training and/or match-play) for male professional soccer players regarding periods of fixture congestion (a minimum of two matches with≤4 days recovery).

Results: A total of eight articles were included in the review. Five studies identified that congested fixture schedules expose players to increased match injury incidence, although layoff duration was typically lower during congested periods. Two studies identified that training and overall injury incidence were higher during congested periods, with another study identifying a lower training injury incidence during congested periods.

Conclusion: Injury risk is, overall, increased during fixture-congested periods; however, the layoff duration is typically shorter. The current findings have implications for practitioners regarding the management, periodization, monitoring, and design of training and competition schedules.

Key Points

- Results suggest that overall (five from the eight included studies), congested fixture schedules expose players to increased match injury incidence, although layoff duration was typically lower during congested periods compared with non-congested periods.
- The data in relation to training and overall injury incidence were somewhat equivocal, with two studies identifying that training and overall injury incidence were higher during congested periods, with another study identifying a higher training injury incidence during non-congested periods and no significant differences in overall injury incidence.

Sports Science Research Update

- Except for injury incidence and layoff duration data, there is a lack of consistency in the reporting of injury data between studies. Fixture congestion is a contemporary and concerning issue (including to the players themselves) and more research is required to add further detail into the injury response observed during congested match schedules.

Sports Science Research Update

The "Worst-Case Scenario": Recovery between repeated high-intensity efforts in Rugby League match-play. Sports Health, Ahead of Print: 1-6, 2022.

Background: Repeated high-intensity effort (RHIE) activity is known to be a critical component of high-intensity, intermittent team sports. Despite the importance of this quality, there are few studies comparing the RHIE activity of higher and lower-ranked teams competing in the same competition. This study characterized the distribution of recovery times between RHIE in Top 4 and Bottom 4 semi-elite rugby league players.

Hypothesis: Players from Top 4 teams will engage in a greater frequency of RHIE bouts than Bottom 4 teams, with shorter recovery periods between consecutive efforts.

Study Design: Cohort study.

Methods: A total of 104 semi-elite rugby league players from 11 teams wore global positioning system units during 28 Queensland Cup rugby league matches. Recovery between efforts was classified as <10, 11 to 20, 21 to 30, 31 to 60, and 61 to 120 seconds.

Results: The majority of RHIE bouts were performed with ≤10 seconds recovery between efforts for both Top 4 and Bottom 4 teams. Top 4 teams performed a greater proportion of RHIE bouts with ≤10 seconds recovery between efforts than Bottom 4 teams (effect size [ES], 0.27 ± 0.19), while only trivial differences (ES, 0.17 ± 0.20) were found between Top 4 and Bottom 4 teams for the proportion of RHIE with 61 to 120 seconds recovery between efforts. Wide-running forwards from Top 4 teams performed a greater proportion of RHIE bouts with ≤10 seconds recovery between efforts (ES, 0.59 ± 0.40) and a smaller proportion of RHIE bouts with 61 to 120 seconds recovery between efforts (ES, 0.62 ± 0.38) than wide-running forwards from Bottom 4 teams.

Conclusion: Rugby league players perform proportionally more RHIE bouts with ≤10 seconds recovery between efforts than any other recovery duration. In addition, Top 4 teams are more likely to complete a greater proportion of short (≤10 seconds) recovery RHIE bouts than Bottom 4 teams.

Clinical Relevance: These findings demonstrate the demanding nature of RHIE activity in rugby league. To be successful, teams should train for these demanding passages of play.